

***Talent and The Social Physics of Inequalities*¹**

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Abstract

Talent and creativity are widely regarded as the ultimate factors and rationales for economic success in societies experiencing rising income inequalities, job polarization, global mobility of high-skilled workforce, as well as long-standing admiration for undertakings that combine outcome uncertainty and highly individualized ascription of performance. Yet, due to substantial definitional issues, the content and causal effectiveness of both factors remain obscure, if not dismissed as sheer social constructions. Drawing mainly on studies of artistic and scientific work, the paper offers cues to solve the puzzle, by relating four research strands: the disproportionate gaps in creative career trajectories arising from relative, performance-based valuation; the multidimensionality of skills and the multistage technology of their formation; the variability of non-routine work; the stratification effect of assortative matching in teamwork and collaboration. The final section examines how the workforce “talentification” impacts compensation practices, with pay scales getting legitimized according to large, supposedly Paretian-like individual performance differentials. The conclusion discusses this impact.

Le talent et la créativité sont considérés par beaucoup comme les facteurs et les justifications ultimes de la réussite économique dans les sociétés qui sont confrontées à des inégalités de revenus croissantes, à la polarisation des emplois, à la mobilité mondiale de la main-d'œuvre hautement qualifiée. Nos sociétés valorisent aussi, depuis longtemps, les activités qui associent l'incertitude des résultats à l'attribution hautement individualisée de la performance. Pourtant, quand il s'agit de définir le talent et la créativité, les problèmes sont nombreux, ce qui contribue à rendre obscurs le contenu et l'efficacité causale de ces deux facteurs, ou même à y voir de pures constructions sociales. Prenant appui principalement sur des études du travail artistique et scientifique, l'article propose des pistes pour résoudre l'énigme de la diffusion universelle d'une notion malaisée à définir, en reliant quatre axes de recherche : les écarts disproportionnés dans les trajectoires de carrière créative qui ont leur origine dans l'évaluation relative des performances individuelles ; la multidimensionnalité des compétences et la technologie séquentielle de leur formation ; la variabilité du travail non routinier ; l'effet de stratification de l'appariement assortatif dans le travail d'équipe et la collaboration. Dans la dernière partie sont examinés les effets de la « talentification de la main-d'œuvre » sur les pratiques de rémunération des salariés et les arguments pour légitimer des écarts

¹ The present paper draws on research and analyses that were presented first at the 2017 Annual Meeting of the American Sociological Association and at the conference I organized at the Collège de France "Le talent et ses approches en sciences sociales. Attributions, Marchés, Mobilités, with partial publication in the book I edited on *Le talent en débat* (Paris, PUF, 2018). To revise and extend it, I then benefited from numerous exchanges with sociologists and economists at lectures and conferences in Paris, Strasbourg, Chicago, Brussels and Lausanne in 2019, 2020 and 2022.

supposés résulter d'inégalités parétiennes de performance individuelle. La conclusion discute ces arguments.

Keywords : talent, stratification, assortative matching, compensation, inequality, art, science, business management.

For several centuries, the talent semantics has been used to qualify outstanding performances in specific fields of human endeavor and foremost in the arts, scientific research, R&D, sports, and entrepreneurship. In these fields, the measurement of individual performances is an ordinary cog of competition, ascending selection in tournaments and the rating industry, whether the issue is the determination of market value of artists and athletes, the justification of recruitment, promotion or poaching in the academic world, or the skewed allocation of distinctions and prizes. The tests during which the success of the more talented is obtained by direct competition are the very essence of sporting shows, and the same is true for the popular industry of talent shows. In classical music, aspiring concert artists feed in growing numbers the bubbling industry of international competitions (McCormick, 2015, Menger 2023).

The vocabulary of talent, genius or brilliance is the usual means to ascribe exceptional qualities in activities where success is hard to predict (Stinchcombe, 1968). This terminology marks a clear difference with all the other usually measurable qualities of an individual. They are essentially attributed *a posteriori*, because the objective is to encompass and categorize what cannot be predicted, but only “retro-dicted”, in the light of experience. Uncertainty is then converted into a promise of enduring success. This explains why talent and reputation are knit together, since the reputation mechanism operates quite in the same way, according to game theory. Paraphrasing the German sociologist Niklas Luhmann, one can state that the attribution of talent acts as a mechanism of complexity reduction for the evaluation of people.

In the 18th Century, American and French revolutionaries defined legitimate inequalities by two qualities: virtue and talent. Article VI of the French Declaration of the Rights of Man and the Citizen (1789) stipulated that “all citizens, being equal [in the eyes of the law], are equally eligible for all public ranks, positions, and employments in accordance with their capability and without distinction except in regard to their virtues and talents.” Virtue referred to individual behavior, or effort and dedication to work, and talent to natural endowment.

In order for an unequal distribution of « natural » endowment to get legitimate, a series of conditions had to be met (Rosanvallon, 2013, Menger, 2017)

- the notion of talent remained poorly defined as a natural disposition, an aptitude for certain things, a capacity, an ability ; yet it invariably referred to eminent qualities, and outstanding capabilities
- talent referred to an uncorrupted new order, as opposed to the old society of orders and inherited privileges, an order of democratic liberal society, whose pillar was to be the emancipation and self-realization of the individual ;
- though being the signature of eminent singularity, and being unequally distributed by nature, talent referred to the emancipatory order of a positive sum game : one's talent flourishing should be viewed as a road to successfully join the social, cultural or economic elites. Moreover, striving to excel would be turned into a positive sum social game, as far as excellence could be understood and imposed as a moral dedication to one's self-realization.

Elites started to get democratized. Careers open to « talents » in the army (Bien, 1979; Blaufarb, 2002), in business and entrepreneurship and in the arts and sciences then symbolized the power of individual merit, as a combination of well-committed effort (« virtue ») and talent.

As mentioned, certain domains were ideally suited to the flourishing of talent, and most prominently the arts and sciences. At the time talent became the pillar of a new merit-based society, originality went on to get celebrated as the founding principle of self-realization in those domains allowing for full deployment of talent - the arts and sciences.

The matrix is the coupling of individualization and eminence that lies at the historic root of the identification and valuation of talent. On the one hand, individualization may well refer to a process of non competitive, originality-driven differentiation, of unhindered actualization of one's capabilities or abilities. On the other hand, eminence and hierarchy have inhered in the semantics of talent since the parable of talent has been invented, discussed and interpreted at length by generations of theologians from the early centuries AD on (Mombello, 1976). If so, talent may refer to the way difference is converted into hierarchy and inequality.

Over the last fifteen years, since the “war for talent” label was coined by McKinsey and flagged by the whole community of HR management executives and consultants, the notion of talent became more and more widespread in organizations. Nowadays, the analytical measurement of individual performance and value feeds an emerging technology called people or talent analytics (Isson and Harriott, 2016). The talent market, and the talent traffic as they call it in Silicon Valley, have developed on a global scale to include “talent flows” of highly skilled migrant workers (Kerr *et al.* 2016).

Since the vocabulary of talent has spread out across organizations and businesses, the lamentations on the impossibility of defining talent has become as frequent as is intense the incorporation of this notion in the human resources management of the organizations. On the one hand, as shown in innumerable literature reviews (for a recent convincing one, see Cappelli and Keller, 2014), everyone complains that there is no consistent approach (Gallardo-Gallardo *et al.* 2013)

On the other hand, talent management has indeed become a key concern for firms and their leaders and executives. The literature, whether academic or not, is burgeoning (Dries, 2013). It thus may well be that « talent is just one of many slippery HR buzzwords that have wormed their way into our collective consciousness ». (Briner, 2015)

The notion of talent falls within a wide range of concepts such as “potential”, “abilities”, “high-skills”, or even “being a graduate of one of the most highly selective universities”, or *grandes écoles* alumni in France (Vallée and Célérier, 2015). Therefore, a lot of research in economics, management, psychology, educational sciences tend to treat those concepts as interchangeable, by arguing that a notion as holistic as talent can only be approximated through indexes or characteristics we know to be imperfect – in other words, by proxies.

This principle of individualized attribution of qualities so difficult to define is paired with highly skewed distribution of rewards. The winners grab the bulk of the circulating rewards

(monetary and non-monetary). No wonder, the adoption and celebration of the talent currency correlates with unprecedented levels of inequality in income (Atkinson and Bourguignon, 2000; Frank and Cook 1995; Kaplan and Rauh, 2013; Keister, 2014). At the same time, the institution of such personal value celebration is elevated to the rank of admirable success when it is supported by a remarkable property of creative industries, the test of time. As a matter of fact, the time scales of success celebration can go from the shortest term (the views and likes on a website where a given content has been posted) to the longest term (the indefinite horizon of admiration granted to celebrities in arts and sciences). This constitutes the unique frame of the transformation of success' contingencies into a "heritage-worthy" capital, that is to say into works and personalities given as models.

We can take stock of the characteristics of talent identification and exploitation in the arts and sciences. These sectors are virtuoso in the pairing of multiple competition formats and multiple excellence celebration institutions and formats. Furthermore, they are expert at converting inequalities of success into motives of admiration, more than envious comparison, at least from the consumers' side. And finally, they are expert at over-exploiting the professional and lay audiences sensitivity to differences in quality in individuals and their achievements. The question that comes is : are those characteristics the ideal vector for the diffusion of the notion of talent towards the general labor market, firms and organizations?

A roadmap

My aim in this paper is to follow another route, by taking seriously the current "talentification" of the personnel management and its secular discontents. The seeds of these discontents lie in the fact that the presumption of self-actualization at work contained in the "talentification" principle runs against the highly unequal chances of achievement that are observed in high-skilled, creative occupations. Indeed, as the functioning of core creative worlds such as the arts and sciences shows, one crucial issue is missing: that of the several dimensions of inequality magnified by the work system which builds on highly individualized performance ratings and selective matchings.

To take the notion of talent seriously, I shall carry out a theoretical analysis in order to avoid two well-known pitfalls, the essentialist path that turns retrodiction in a tautological determinist description, on the one hand, and constructionist criticism and its several variants, on the other. Drawing on studies of artistic and scientific careers, the present paper brings together four research streams that respectively deal with: the disproportionate gaps in creative career trajectories arising from relative, performance-based valuation; the multidimensionality of skills and the multistage technology of their formation; the variability of non-routine work; the stratification effect of assortative matching in teamwork and collaboration. The paper thus aims at bridging the gap between a dispositional and a processual view of creative undertakings and achievements, by obsessively looking for decomposability.

First, contrary to the previous, demand-based approach, a supply side model may well start with negligible ability differences and end up with enormous gaps in success that are ascribed to a mysterious talent differential. Two causal candidates are discussed, on the demand and on the supply side. The cumulative advantage model endogenizes presumed differences in

talent. The human capital model gives room to unobservables also called abilities and residuals, that are either exogenous signatures of individual heterogeneity but also to unobservable random shocks and events. It can be checked whether random factors should suffice to trigger a cumulative advantage that results in achievement differentials.

Second, instead of erasing individual heterogeneity, I'll open the black box, and redefine talent as the holistic label for a multifactorial set of determinants interacting in a multiplicative way. Support to the resulting model comes from various strands of research.

Third, recast along those lines, the understanding of talent needs a further specification : which domains and jobs do call upon and benefit from the multiplicative interaction of abilities? Non-routine jobs that allow these abilities to have a multiplicative effect on an overall outcome of a project are obvious candidates. One may suspect that invoking "non-routine", the new flag of job stratification, merely amounts to substitute a new black box on an older one. Yet, substituting effort and motivation for ability raises the same objection.

Fourth, while talent stands at the peak of the individualized ascription of performance, the model I suggest needs the assortative matching mechanics as an essential complement.

In the last section, I claim that while the pendulum in the talent management literature and consultancy business seems to swing between the inclusive and exclusive version of talent, the latter is in fact the dominant approach. A superexclusive valuation of talent has even emerged, that tries to legitimize overtly Paretian pay-for-performance schemes of compensation in various firms and sectors.

The conclusion provides avenues for taking another view.

From the demand side to the supply side : talent as a driver and a residual

The demand for quality being a core dimension of valuation, perceived differences in quality are of decisive importance in orienting the consumers' preferences. At a given price for a good or service, a consumer's utility will be greater if she chooses a professional who is considered more talented than another. This seems quite obvious when objective quality superiority is at stake. A surgeon with a 10% greater ability to save lives than others will be in greater demand. Inequality in reputation and earnings is then based on performance differences and non linear increase in expected utility from higher quality (Goode, 1979 ; Rosen, 1981). In the arts, perceived talent and commodity quality differentials point to subjective utility : a quality difference that yields greater subjective utility is an inherent feature of the service demanded by the audience. Originality-driven production widens the scope for individual differentiation and leads to a more convoluted and noisy process of evaluation, as it leans toward the subjective side of perceived quality.

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increase in expected utility from higher quality (Goode, 1979 ; Rosen, 1981). In many non-routine services, perceived talent and product quality differentials point to subjective utility.

Professionals work to differentiate themselves from one another on multiple levels so as to make their mark in the competition. Measurement is very often in relative terms : evaluations are based on incessant comparisons. What is called “talent” can be defined as the quality gradient attributed to the individual worker through these relative comparisons. The difficulty in defining talent rests in that it is not an arbitrary value, but rather a purely differential quality (Menger, 2014).

Ordinal rankings generated by comparisons leave the magnitude of that differential quality indeterminate. According to Rosen (1981), even minimal perceived differences in quality among professionals can suffice to concentrate disproportionate demand on those deemed either significantly or slightly more talented than others and to win them (monetary and/or symbolic) rewards, a reputation and opportunities for working that will greatly enhance and bolster their competitive edge. Moreover, as Rosenbaum (1989) has underlined, comparative rankings do not merely reveal unequally distributed qualities (of whatever magnitude differences are), and then select individuals on that basis. These competitions cause divergences in contenders’ career paths, despite the fact that their abilities may have been rather similar. The concentration of earnings and reputations on a very small number of individuals could therefore correspond to gaps in success that are far greater than the gaps in abilities known as talents. The resulting disproportion wouldn’t operate if sensitivity to quality differences didn’t exist, if attention weren’t scarce and if informational constraints could vanish.

To sum up, viewed from the demand side, talent is a matter of differential coupled to sensitivity to quality differentials. Imperfect substitutability between workers, as revealed by peers, intermediaries and consumers’ preferences, gives rise to differences in demand that favor those deemed most talented among them. Rosen’s model shows how differences in earnings can be disproportionate to differences in talent, yet it posits that differences in talent do exist, and that demand is sensitive to those differences. Disproportion means that differences in talent may be tiny or “negligible”, or possibly considerable : in any case, they exist and can be perceived, even if imperfectly or thanks to enormous expertise. This amounts to what a social constructionist might call minimal essentialism.

Quite in the same league, yet on the supply side, lies the Mertonian model of cumulative advantage, which derives from the Matthew parable of talent. Note that the demand-based model contains its own mechanism of cumulative advantage, based on rational contagion of consumers or professional audiences (Gould, 2002), the link between the supply and the demand side in the cumulative advantage argument being provided by the selective attention of professionals acting as consumers vs. co-workers. The question is still the same : what does the quality difference that triggers the magnification process consist of? If abilities were readily definable or observable, there would be no uncertainty about performance and success : a system of perfect screening to detect them as early as possible and of perfect competition to incentivize them and get the highest returns would operate. But if things did happen that way, the factorial breakdown of causes of inequalities in rewards and careers should be able to capture the contribution of determinants as strong as abilities and highly

unequal endowments. This is precisely what earnings equations cannot do.

A multidimensional and multistage production of ability

Basically, human capital theory explains those aspects of productivity and earnings that result from investment in formal education, further training and accumulated job experience. These are straightforwardly observable earnings determinants. Yet as Mortensen (2003, p.1) reports, “Although hundreds if not thousands of empirical studies that estimate so-called human capital wage equations verify that worker characteristics that one could view as indicators of labor productivity are positively related to wages earned, the theory is woefully incomplete in its explanatory power. Observable worker characteristics that are supposed to account for productivity differences typically explain no more than 30 percent of the variation in compensation.”

Thus less observable or unobservable factors are commonly invoked to explain not only why earnings greatly differ across groups of workers defined by occupation, education, experience and other observed traits, but also why earnings vary within groups that are observationally similar. Such unobservables as talents, abilities and whatever name is given to residuals describing disparities among persons are commonly invoked to explain productivity and pay dispersion among seemingly similar workers (Atkinson and Bourguignon, 2000). These unobservables are the signature of individual heterogeneity: rather than resulting from investment or accumulation, they seem to qualify the a priori distribution of non-producible endowments. Yet among the unobservables, we shall also count random shocks and events. If workers were similarly endowed at the outset, and some factor intervened to generate inequality, due to a cumulative advantage process, performance and reputation differences might be entirely disconnected from talent.

The mechanism of cumulative advantage is activated as soon as a performance gap appears among a set of candidates for success, and as soon as one of the young scientists obtains an advantage over others. The explanation lies in the process of self-reinforcement: Recognized for a remarkable performance early in his career, the scientist attracts the attention of his peers and receives support from mentors and colleagues who are more advanced in their careers. It is thus easier—i.e., less costly—for a potentially high-status researcher to produce work of a given quality level by getting readier access to various resources (funding, research time, collaborations) and to increase his chances of enlarging his audience (Merton, 1973; Podolny, 2005).

It is thus understood that what causally determines the early attainment of a strong reputation and a high status is the difference in value between the achievements of competing scientists in the same age group. And it is also clear that this causal link reinforces itself as advantages accumulate—to the point where what may have been a slight gap in performance at the outset comes to appear as a significant, intrinsic difference in quality. As DiPrete and Eirich (2006) point out, Merton’s cumulative advantage model leaves open the question of differences in talent as a source of corresponding differences in performance: This causal origin might be an intrinsic difference in ability or quality (however one defines quality or the combination of superior qualities possessed by the most brilliant competitor), or simply a random factor. By lowering the initial difference to nothing or nearly nothing, Merton aims to dismiss the argument that would reduce the analysis of cumulative advantage to a simple

tautology.

Should the model go so far as to entirely erase individual heterogeneity and locate the causal kick out of the people's control ? In that case, a constructionist model of inequality like the Marxian one would apply : homogeneity of work and workers, quality as a multiple of quantity, equal endowment in multiple talents, constraints originating in the prevailing social and economic organization of the labor process and allocation of property rights, scarcity understood as a process rather than a law and limit. In a more realistic scenario, a homogeneization parameter can be used to level the playing field : as pointed out by Allison and Stewart (1974) for scientists, those workers already form such a rigorously selected elite that we expect them to be quite homogeneous on several essential dimensions predicting occupational achievement.

Even then there is much left unexplained. Two possible avenues suggest themselves for research to go further. The cumulative advantage process may have operated much earlier and people that seem similarly endowed, at some starting point of the professional career, follow in fact radically different paths. Alternatively, underlying differences in domain-specific abilities, motivation or some other behavioral determinant of productivity exist whose impact becomes apparent and efficient. In actuality, a unifying model may account for both interpretations.

The model has the following elements : 1) Ability is multifactorial : it is better conceived as a bundle of characteristics and dimensions rather than as a unique endowment magically labeled talent 2) the determinants interact in a multiplicative way ; 3) Determinants of productivity and success in performing non-routine tasks are more numerous than in routine ones; 4) Some of those determinants are observable, others hardly so, partly because they reveal themselves and develop only processually, from experience and on-the-job learning, given that non-routine work and higher variability in task performance result in a higher learning content of work, that gradually provides its performer with essential information about the quality of his job-match; 5) assortative matching are a major cog in the multiplicative function of production.

Let me quote an example put forward by various authors (Shockley, 1957 ; Allison and Stewart, 1974 ; Neal and Rosen, 2000), to break down the all-purpose factor of talent when considering the highly skewed distribution of productivity and rewards in high skilled occupations. How do we explain inequalities in research productivity? A first approach transforms scientific activity into a normal activity in which the gaps in success levels are explained by a combination of factors that are usually responsible for differences and variations of performance in high-skilled jobs: social background, age, gender, quality of education, working conditions, and the organizational environment. Attempts to measure one by one the correlations between those human capital factors, on the one hand, and scientific productivity, on the other hand, always produce disappointing results: correlations are weak or inconclusive.

Now consider productivity in the following way: Take the propensity to publish papers, as discussed in Shockley's paper. What factors may be responsible for the unequal production of

articles? Shockley proposes a list, to characterize the process leading to the publication of an article, a process that amounts to a series of tests:

1. the ability to find a good problem to explore,
2. the ability to explore it,
3. the ability to recognize a fertile result when it appears,
4. the ability to know when to stop and to write up results,
5. the ability to write one's article well,
6. the ability to learn from criticism (to be constructive rather than defensive),
7. the determination to submit one's article to a scientific journal,
8. perseverance to make changes and to react to the observations of the journal's referees.

The propensity to publish is then the result of a log-normal distribution, due to the interaction between numerous independent factors, each of which is distributed normally. Because they must be combined, they determine productivity in a multiplicative rather than in an additive way. In other words, a set of normal causes, by composition, produces unusual results. The resulting distribution tends towards a log-normal distribution profile: as the number of factors involved in the multiplicative function of production increases, so does the skewness of the distribution.

More generally, research productivity and creative inventiveness may be modeled as the outcome of a number of features that interact multiplicatively rather than additively. For example, the model may include personality traits that are salient from studies in social and cognitive psychology of creative behavior and research productivity: ambition, perseverance, resilience, aversion to vague situations and to fuzzy compromises, a high coefficient of strong independence, the fact of being demanding of others, intrinsic motivation for obsession-driven work, and a taste for constant experimentation.

At the root of the multidimensional characterization of abilities lies the distinction between cognitive and non-cognitive traits. Three Centuries ago, the Enlightenment celebration of talent as a driver of democratic occupational achievement and social mobility was already based on the synergistic coupling of talent and effort, the former being a mere capacity in need of a powerful disposition to act and persevere in order to get fully actualized. After reviewing the psychological literature on talent and genius produced since 120 years, Eysenck puts at the center of his analytical construction the multiplicative effect of the numerous determinants (cognitive, environmental and personality variables) that act synergistically, to account for creative achievement and consistent outstanding performance (Eysenck, 1995; see also Simonton, 2001).

In a way, yet a very twisted one, Bourdieu's attempt to explain how innovations by major artists can be traced back to their unclassifiable social identity (Bourdieu, 1996) has a flavor of what is investigated here as a multiplicative function. In Bourdieu's view, the individual of the highest talent is a statistically improbable figure because he combines in himself characteristics that are usually separate. But the combination of characteristics that make for the innovator's endowment seems quite ad hoc. Bourdieu's insight hardly escapes a contradiction commonly observed in sociological writings on great innovators, that of the epistemological doubling of the explanation. On the one hand, according to determinist

analytical schemas, although the rank-and-file artists' calling and career are presented as controlled by market or social forces, the outstanding artist's behavioral oddities become one of the indices of her social unclassifiability. On the other hand, a strategic interpretation of the creator's behavior and psychology tends to make him an ambitious, calculating figure, determined and rational, who is capable of successful maneuvering.

Recent research literature on the economics of human development and early educational investment programs (e.g. Cunha and Heckman, 2007) that focuses on ability and development follows the same paths. It posits that : 1) agents possess a "vector of abilities that range from pure cognitive abilities (e.g., IQ) to noncognitive abilities (patience, self control, temperament, risk aversion, and time preference), and are used with different weights in different tasks in the labor market and in social life more generally"; 2) ability gaps between individuals and across socioeconomic groups open up at early ages, for both cognitive and noncognitive skills, due to a multistage technology which is based on dynamic complementarity and self-productivity that have multiplier effects: "skills beget skills and abilities beget abilities". Skills are self-reinforcing and cross fertilizing; 3) interaction of nature and nurture in producing abilities and behaviors is intricate, instead of relying on linear additive models.

The main analytical challenge to take up remains the specification of those determinants in interaction, as decomposability has a long way to go.

The nature of tasks and jobs: routine and non-routine

Repeatedly, in controversies about inclusive vs. exclusive identification and management of talent, the question arises as to the nature of tasks and jobs that call for abilities beyond the standard human capital investment in skill acquisition. Along with the rising use of the talent currency in personnel management, a revival of the functional stratification theory of jobs has supported a fine-grained analysis of tasks, according to their routine vs. non-routine content, since the distinction between routine and non-routine jobs has gained enormous attraction in the literature of job polarization and skill-biased technical change (Acemoglu and Autor, 2011).

By characterizing determining factors in terms of their variance, we can map activities along an axis running from the most standardized and repetitive to the least routine, based on whether the determinants of action have low or high variance. Artistic creation and scientific research, but also less prestigious activities such as advertising, gambling, fighting, sports, and the stock market, figure among the least routine of human enterprises, and their outcome is imperfectly predictable. For sure, production in those worlds would be impossible if there were no conventions or routines (Becker, 1982) to allow for specialization and cooperation among the various categories of personnel involved. Yet the fact remains that the seductive power and the very prestige of creative professions are related to their degree of unpredictability of outcome and success.

In organizational terms, the distinction between routine and non-routine tasks leads to the strategic differentiation of jobs, which is at the heart of the current talent management literature (Cappelli, Keller, 2014). It echoes a famous functional differentiation, put forward by Stinchcombe fifty years ago, in order to connect a dual characterization of jobs or

‘industries’ and the rate of variability in performance and outcome. With subsequent refinements and elaborations of the model in mind (Stinchcombe and Harris, 1969 ; Jacobs, 1981 ; Kremer, 1993 ; Baron and Kreps, 1999), we can summarize the reasoning as follows. Both jobs and skills are to be defined in the same probabilistic way. Skill is the probability of performing the task perfectly vs. badly. Jobs differ in the success rate of their incumbents : in some positions, success is infrequent and outstanding achievement exceeds the norm by far, whereas in other skilled jobs, normally trained workers rarely fail, which leaves little room for superlative performance and its appreciation. To locate where the superstar economy and its corresponding talent chasing occur, Rosen (1981) notes :

« Some tasks are so routine and so circumscribed by existing practice that nearly any competent person achieves about the same outcome. Others are more difficult, more uncertain, and, this being so, allow greater possibilities for alternative courses of action and decision. Such tasks offer greater scope for superior talent to stand out and make its mark. More capable physicians spend smaller fractions of their time on routine cases and larger fractions on difficult ones than do physicians of more modest ability »

When applied to the strategic dimension of jobs within the organization, the argument, in its Stinchcombe-Jacobs-Baron-Kreps version, states that the first category of jobs includes activities in which the value of an exemplary performance has a multiplicative effect on the overall performance of the project or the organization, even if the contribution to the success of the project or organization may well be out of proportion with the difference between the worker’s abilities and those of his partners. In such activities, there is a fierce competition to attract and reward individuals deemed exceptionally talented: concentration of rewards (monetary and/or non monetary) creates situations of winner-take-all or winner-take-the-most. Stinchcombe mentioned the following activities and "industries": scientific research, the entertainment industry, and sports. Talent in these sectors or professions is a “complementary” or “multiplicative” factor of production.

The statistical expression of a winner-take-the-most situation is impressive in science : individual research productivity (in terms of publications) follows a Pareto distribution, and the distribution of citations of papers even more so : one-fifth of all papers published in a given field earn over 80% of the total of citations given (Seglen, 1992; Larivière *et al.* 2010). In the arts, income distribution also follows the Pareto curve: one-tenth of professionals in the given field earn half of all annually distributed income; one-fifth earn 80%. And there are more individuals earning nothing—or less than nothing after art-related expenses deduction—than in any other higher occupation (Alper and Wassall, 2006). This means that people are also attracted to such occupations on a basis of non-monetary rewards, or so-called « psychic gratifications ».

The second category consists of those positions where individual contributions—even outstanding—cannot considerably increase the organization’s reputation or profit. In these activities, the required skills constitute an “additive” factor of production, and they are more normally distributed among the individuals concerned. Teaching, manufacturing, or craft production are occupations wherein the presence of professionals who display exceptional (or deplorable) performance will have little effect on the total performance of the organization.

Take again the example of academic work and employment, and the link between the core tasks of teaching and research. In research, the distribution of individual productivity and professional visibility has a highly skewed, Pareto-like profile, whereas individual performance in teaching has a normal, Gaussian, distribution. The production function of teaching is additive, while that of research is multiplicative. This is why management of research activities has granted increasing importance to assortative matching and the concentration of critical masses of talent to leverage the faculty's research potential. Meanwhile, teaching staff becomes more substitutable once they move (or are moved) away from the frontiers of advanced research. Unsurprisingly, given the crucial importance of reputational capital to higher education institutions, tension between research and teaching missions has mounted (Menger, 2016).

The production of outstanding performance in high skilled activities in which the odds of a superior achievement are low corresponds to what Simon (1962) calls a complex system that takes the form of a hierarchy. Performers are ranked according to their relative level, and move up to higher strata as they are willing and able to improve their performance. Performance in each stratum is made up of a large number of factors that interact in a non-simple way, while stratification results from the fact that the identification and interaction of those various determinants are subject to a tighter organizational control as one moves up to the upper levels. And, according to the Simon model of complexity, each determinant of the performance (technique, motivation, attitude, etc.) has to be itself modeled as a subsystem containing various determinants in interaction. After all, motivation and effort are anything but a simple additive quantity.

Assortative matching : talent and its selective ecology

As I just mentioned it, the multiplicative function of production that explains the performance ascribed to « talent » contains another salient feature, its organizational structure : beyond one's sense of personal organization, that may count among the abilities to be developed, there is a structure of work organization that amounts to an assortative matching game. This is what the « qualitative jumps » in Chambliss' study refers to, since these jumps result from a « change in settings », that is an improved work organization with better coaches, new training partners, and higher level contests.

Assortative matching characterizes another dimension of the multiplicative nature of the production function. Like a scientist in Merton's cumulative advantage model (Merton, 1968), an artist benefits from associating with professionals reputed to be of equal or superior quality in their respective careers. Indeed, in order for an artist to secure the best chances of developing his talent, it is important for him to associate with professionals of comparable value working in the other professions involved in the production and circulation of artworks

Furthermore, in the very early stages of artistic and scientific careers, formal learning trajectories followed by on-the-job learning (and even by apprenticeship, a persistent form of artisanal organization in certain artistic and technical-artistic trades) are heavily determined by an individual's association with experienced partners. These are the people who provide an artist on the path toward professionalization with better opportunities for

developing his skills in demanding projects, through contact with fellow artists who have themselves been selected based on their potential. Indeed, the unique characteristic of assortative matching is that it provides “matched” individuals with higher returns on their respective abilities than they would otherwise obtain in the case of random matching.

Art worlds combine labile organizational architectures (e.g., network, project, vertical disintegration) with teams structured through the association of professionals of equivalent quality or reputation—that is, through assortative matching. Labor markets for the most highly qualified jobs are, in this way, hierarchically ordered by professional pairings. A successful career dynamic constitutes a movement of upward mobility in a world stratified by networks of acquaintances and recurring collaborations. Assembling professionals of approximately equal level—each in their respective function has a multiplicative effect on a given project’s chances for success and on the project collaborators’ chances of accumulating reputation (Accominotti, forthcoming; Kremer, 1993 ; Rosen, 1983). The same is true of hospitals, law firms or universities. Professional tasks are both subject to multiplicative quality interaction and to learning spillovers.

The academic world builds with unprecedented strength on the same assortative game in order to come up with the challenges posed by the rat race of research on global scale. In a framework of positional competition, top universities not only lead the field by attracting the best scholars and selecting the best undergraduate and graduate students on a global scale, but also impose their costly, research-based educational technology on the followers (Menger *et al.*, 2015). The magnification of inequality in research has to do with the fact that research is open to the broadest environment possible (Waltman *et al.* 2011), whereas teaching is a service confined to an organization. Some of the most significant productivity gains stem from the team organization of work in research (Wuchty *et al.*, 2007) and from the networked architecture that follows the rules of assortative matching. Learning spillovers increase as high quality teaching and research are matched, since high quality students and academics are better able to teach and learn from their peers (Winston, 1996).

The mechanism is intriguing. On one hand, it works to reduce uncertainty in that it enables us to explain how, after a series of career stages organized by comparison tournaments, a hierarchy gets established that in turn structures collaboration among professionals. The idea here is that individual quality which could not be identified at the outset ultimately reveals itself. On the other hand, the same mechanism works to increase the individual’s chances of accumulating reputation as well as new skills, in that it facilitates rewarding and productive collaborations, which might suggest that certain abilities emerge that did not previously exist. Indeed, the quality of professional matching acts like a signal informing others of the individual’s reputation, but it also affects that individual’s chances of accumulating new skills and enhancing her prospects by integrating her into a collaboration and project environment that increases the formative potential of her work.

The selective matching argument seems to come close to Bourdieu’s structural model of positional homologies but in fact, assortativeness is not merely a matter of managing positional signals. The selective job-matching argument is not characterized by the same stasis as that imposed by structural homologies. For example, it may be optimal for a given individual to become involved in a project that leads her to collaborate with partners who enjoy much

less recognition or status than she because that project is deemed more innovative, risky and formative than more predictable collaboration arrangements. Without this dynamic, we could not understand how careers get coupled with innovation or how diversity gets introduced into a stratified system. But without the analysis of hierarchies and their structuring character, we could not understand selective co-optation mechanisms. In sum, if those mechanisms were the only ones operative they would dry up activity, because bringing together the best professionals in every field is not enough to guarantee innovative work.

Talent and compensation practices. How to legitimate extreme inequalities?

The talent management approach that has spread in companies and organizations from the late 1990's on was based on a simple thinking : talented people perform so much better than the company able to employ and develop the most talented people will succeed. How to attract talents, how to keep them and how to build a talent pipeline of young high potentials that are to be found, developed and retained are therefore the key issues. Then comes the question of definition and identification. The parable of the talents is meritocratic and inegalitarian. On the one hand, as we each have strengths, the best that talent managers have to do is to uncover and leverage them. On the other hand, the parable suggests that some people are endowed with greater abilities than others, get greater success and should secure greater rewards. The pendulum then swings between the inclusive and exclusive version of talent in the talent management literature and consultancy business, whereas everyone puzzles over the definitional maze. As noted by Cappelli and Keller (2014), insofar as the investment of scarce resources is at stake, the dominant approach of talent in firms is now clearly the exclusive one. To differentially invest in certain individuals and jobs is in line with resource optimization in the organizational strategy literature as well as with the 'Matthew effect' model of cumulative dynamic differentiation among people.

Within that dominant, exclusive framework, the next question is : where should we locate the causal origin of the workforce differentiation ? Half a century ago, Galbraith claimed, in his book *The New Industrial State* (1967), that the success of the firm depended more on the quality of its organization and the pertinence of its management procedures than on the exceptional talents of this or that individual.

“The real accomplishment of modern science and technology consists in taking quite ordinary men, in forming them narrowly and deeply, and then, through appropriate organization, arranging to have their knowledge combined with that of other specialized but equally ordinary men. This dispenses with the need for genius. The resulting performance, though less inspiring, is far more predictable.” (Galbraith, 2007[1967], p.75-76)

As noted by Rosanvallon (2013) in his comment, talent was thus taken down from its pedestal. Drucker made exactly the same point. Executives were certainly better paid than workers, but only within the framework of a functional hierarchy of skills. Peter Drucker stated at the same time that the pay ratio between the top executive and the humblest worker should be no greater than 20:1

The current dominant view goes the opposite way, in three main respects. First, the internal

model of career and succession planning to fill executive jobs after sophisticated selection and horizontal and vertical job mobility within the company is in decline. Outside hiring is on the rise, with its trail of puzzling challenges (instilling and incentivizing loyalty to retain those hired and poached high potentials and executives, building efficient teams, avoiding damaging short-termism). The McKinsey report on *The War for talent*, which has been largely seen as a landmark in the rising interest in talent, and in the controversial celebration of a rebranded workforce differentiation, did precisely address those issues in the late 1990's (Michaels *et al.* 2001).

Secondly, the hair-raising escalation of CEO compensation in large firms and leading sectors of the global economy over the last three decades has sparked off protests and indignation, as well as a burgeoning academic literature in economics and business facing the unprecedented challenge of explain those astronomical inequalities.

Thirdly, as pointed out by Cappelli and Keller (2014, p. 324-325), the talent-driven differentiation of the workforce has refocused the attention on the functional characteristics of the jobs : « outside hiring is the dominant approach to meeting talent needs. By definition, outside hiring and executive search put the focus on the job and its requirements, including particular attributes of the organization. Then the process moves to searching for a candidate who might fit those requirements. For current employees, internal job boards, as described above, bypass development and assessment processes when candidates move ». Talent management certainly remains mainly concerned with executive and C-suite jobs, and especially with CEOs' differential abilities. Yet, a more extensive view of 'strategic jobs' has surfaced : « a key insight from the recent workforce differentiation literature is that although executive jobs are almost by definition strategic, strategic jobs can potentially be located anywhere in the organization, depending on the strategic competencies of the organization. (...) That strategic jobs exist outside of the executive suite has been trumpeted as one of the most insightful aspects of this approach » (Cappelli, Keller, 2014, p. 309). Instead of focusing on the individual first, and training and developing him in order to fill various slots of the organization in an ascending and additive model of career and productivity, the talent-driven segmentation of the high skilled labor market indeed shows up as the signature of a new organizational doctrine. Is it merely a trendy way to substitute a black box (« talented people ») with a new one (« strategic jobs ») and a psychology-based focus on individuals with an organization-theory based of personnel management ? By asking how exactly strategic jobs may be defined, Cappelli and Keller warn of an identification problem that sounds similar to the ones raised by the notions and pervading use of 'talent' and 'creativity'².

² « Incorporating the strategic job concept into the general idea of talent management could be straightforward. Rather than simply assuming that only top management jobs are strategic, employers could add other jobs to that list and then decide how best to meet the demand for those jobs: Should they be filled by outside hiring or from within, and if from within, what practices are associated with developing and identifying the appropriate candidates? Talent management obviously involves identifying the requirements of those strategic jobs as well as assessing and developing the abilities of individuals (...) Efforts to operationalize the strategic jobs concept and to empirically identify such jobs within and across organizations, particularly those located outside the executive suite, are sorely needed » (Cappelli and Keller, 2014, p. 310 and p. 324)

A striking consequence of the re-engineered view of talent as assorted with strategic jobs is the legitimization of an astronomical-pay-for-astronomical-performance-in-astronomically-important-jobs compensation scheme. As stated by Bénabou and Tirole (2016), competition for talent leads to an escalation of performance pay and detracts from cooperation within the organization. In finance, in high-tech industries, in corporations, in medicine and in academia, not to mention the sports and creative industries, a growing competition takes advantage of local and global "talent flows". To differentially attract, sort out and retain the most productive performers and to induce them to perform to the best of their abilities, performance pay has become a common device in use. Accordingly, the "strategic-job-assortative-matching" model has been given a prominent role, especially by industries in which talent is highly scalable, due to massive joint consumption technology of their markets.

The scheme and the rationale for it can derive from data and studies on those occupations and "industries" I have been mainly concerned with in this paper. Strikingly enough, these industries and occupations are mainly unincorporated or idiosyncratically incorporated and share a common characteristic : in law firms, in research universities, in sports teams, in artistic project-based organizations and careers, individual performance and productivity in strategic jobs (type 1 "star jobs") are measurable in far less ambiguous ways than in type 2 sectors, occupations and firms. Yet the "strategic job-assortative matching" approach in talent-based management practice makes its way across-the-board, under the umbrella of that extrapolation.

In a series of papers, Aguinis and O'Boyle (2014, 2016) have made a name for themselves by gathering data and statistics about inequality in performance. Their study is based on an aggregate of 229 datasets including "633,876 productivity observations collected from approximately 625,000 individuals" in occupations including research, entertainment, politics, sports and sales. They aim at understanding when the distribution of individual productivity is more likely to follow a power law curve (i.e., distribution with heavier tail). Their explanation lies in the principle of cumulative advantage, with various "features and structures surrounding work" acting as drivers. That the distribution of performance is Paretian, and not Gaussian, and for good organizational reasons, is a resounding success with management consultants, and with a part of the academic community working on talent management.

The conditions under which the assimilation of talents to Paretian contributors to the business production would take off are clearly discernable.

1) The advanced technology sector and the technology intensive jobs, in other words, the knowledge workers, especially in research-intensive and innovation-intensive sectors (R&D, IT, finance, pharmaceutical industry, chemical industry, etc.) are carried to Paretian remuneration schemes.

2) A labor market characterized by intense competition for the most technology-complementary abilities and skills, and by an increasing premium put on creativity and inventiveness can only add an amplifying factor to this evolving architecture of jobs and matchings between individuals and organization, and to anti-poaching and anti-raiding remuneration schemes : this comes to a "War for Talent" act II.

3) Some organizations at the avant-garde of this identification of talent to its artistic or scientific incarnations are precisely specialized in the analysis and treatment of big data, and in the production of algorithms. The development of "Talent Analytics" tools is linked to this evolution, as is the rise of "Sabermetrics" in sports.

Aguinis and O'Boyle excluded from their samples, designed to compute the Paretian distribution of performances, all the individuals whose performance is evaluated by a direct supervisor (in other words, the business employees), as, according to them, such a mechanism is designed to create Gaussian results. In other words, they eliminated all the business employees: a good way to substitute a norm to another, under the appearances of measurement. To take the example of the performance distribution of artists, researchers and sportsmen has an obvious methodological interest. Individualization and performance measurement in those unincorporated professions set a reliable benchmark, at least for the part of the activity directly addressing a demand, a public or a peer community of professionals. Even if nobody in those lines of work is working outside a framework of interaction and interdependence with teams, a staff or an organization, an essential part of the result is attributed to the individual, to his credit or his discredit. To develop talent metrics and talent compensation schemes in line with those of artists and scientists amounts to break through the walls of the organization, and to reduce the complexity of interdependencies to naught, or nearly naught.

Note however that in a deliberately Paretian regime that matches individual productivity measurement to proportionate compensation in strategic jobs, the highest incomes are usually based on multilayered systems of compensation (Kim, Jones, 2017). Income includes bonuses, stock options and deferred capital gains, that are precisely the mark of the individuals' hold upon the whole organization, while a mere salary-based pay for performance would remunerate the top performer as a professional contributor or service provider to the organization. The commonly cited rationale for such compensation scales so is to retain those workers and incentivize their "corporate" loyalty rather than nurturing an army of mercenaries in a rootless talent traffic. To compensate incorporated scalable talent seems to reach the upper limit of the bargaining process, namely the bilateral monopoly game : the negotiating power of the employee correlates with the exit options one's reputation is provided with. No wonder that the talentification of the high skilled workforce nurtures a bonus culture (Bénabou and Tirole, 2016)

In a book dedicated to a very straightforward criticism of some management theses and practices, Pfeffer and Sutton (2006) lash out at forced ranking practices that, on the basis of ratings, rank the individuals according to several predetermined categories, following the analogy flaunted by Jack Welch, the CEO of GE, according to which one could rate and rank the workers as it can be done for the students. As a matter of fact, the reliable traceability and measurement of individual performance is a direct yet far from obvious requirement as far as the functional synergy of activities is a very principle of their incorporation into the organization.

Conclusion

By suggesting that individual performance and work interaction can be decoupled to such an unprecedented way as to legitimate hugely discontinuous compensation schemes, the reasoning seems to boil down to the classical problem addressed by Ronald Coase. Why should there be companies if bilateral market transactions with independent contractors could operate as well, through multiple contractual nexuses assembled according to the needs? In the organizational framework I am discussing, the Coasian questioning mainly applies to the upper-level of highly skilled positions, and echoes studies in sociology that have for long shown how professional identity that translates into having a personal reputation and a market power external to the organization can be negotiated inside the organization (Freidson, 1986; Kanter, 1992). And it relates to studies on contingent highly skilled technical professionals (e.g. Kunda *et al.*, 2002) as well to the "free agency" literature that have gained a new momentum as digital disruption, skill-biased technical change, disintermediation of hiring processes through "talent platforms" and global talent flows have made their way (Boudreau, Jesuthasan and Creelman, 2015). The emerging work models that are thereby celebrated refer, once again, to domains such as sports or moviemaking, where 'loaning talent,' free agency, or assembling project-based production teams is the norm and where talent scalability highly matters.

Empirical evidence has still to be gathered in order to assess to what extent compensation practices come truly close to Paretian pay-for-performance in the above mentioned candidate firms and sectors. The investigation may also be oriented in quite the opposite direction, to give a fresh look at compensation designs that remain more egalitarian than would be warranted by productivity considerations. Here are some avenues to be suggested. First, productivity is but one of the dimensions in which individuals differ : as a source of heterogeneity, social preference may lead agent to "sort themselves between 'selfish jobs' which involve high bonuses but no cooperation among co-workers, (...) and 'cooperative jobs' characterized by muted incentives and cooperative behavior" (Bénabou and Tirole, 2016, 347). Second, as Frank (1984) highlights it, status considerations within organizations can be traded off against compressed pay differentials. Third, in fields in which intrinsic valuation of work is of paramount importance, non monetary rewards build an essential part of the income derived from working in non-routine, mission-oriented jobs : substantial effort may go along with low pay, even after discounting for employment risk and pay variability (Menger, 2014).

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